good general purpose renderer. It's the renderer used in most of this book, so you've already had some experience with it. 3ds Max's scanline renderer is also the most tightly integrated renderer available to 3ds Max, and any feature in the renderer can be connected seamlessly with any other feature in 3ds Max. The renderer supports both scanline and raytracing, so only those parts of the scene that need the extra processing power of raytracing receive it. When 3ds Max Software is the chosen renderer, all its options are controlled through the 3ds Max Software panel in the Render Scene window.

### 5.4.1 Antialiasing

**Area -** Computes antialiasing by using a variable-size area filter.
**Blackman -** A 25-pixel filter that is sharp but without edge enhancement.
**Blend -** A blend between sharp area and Gaussian soften filters.
**Catmull -** Rom A 25-pixel reconstruction filters with a slight edge-enhancement effect.
**Cook Variable -** A general-purpose filter. Values of 1 to 2.5 are sharp; higher values blur the image.
**Cubic -** A 25-pixel blurring filter based on a cubic spline.
**Mitchell-Netravali -** Two-parameter filter; a trade-off of blurring, ringing, and anisotropy. Setting the ringing value higher than 0.5 will impact the alpha channel of the image.
**Plate Match/MAX R2 -** Uses the 3ds Max 2 method (no map filtering) to match camera and screen maps or matte/shadow elements to an unfiltered background image.
**Quadratic -** A 9-pixel blurring filter based on a quadratic spline.
**Sharp Quadratic -** A sharp 9-pixel reconstruction filter.
**Soften -** An adjustable Gaussian softening filter for mild blurring.
**Video -** A 25-pixel blurring filter optimized for video applications.

### 5.4.2 Raytracer

Raytracing in the scanline renderer allows it to create highly realistic reflections and refractions. Raytracing works by tracing rays of light from the camera throughout the scene. Raytracing is configured in two places. First, the material must be a Raytrace or another type of material with Raytrace maps in the reflection or refraction channels. Second, the renderer must have raytracing enabled. The raytracing tab is in the Render Scene window.


Paint raytracer